Keiwan Soltani

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Summary

Computer Science Ph.D. candidate with research and industry experience in artificial intelligence, machine learning, UAV and IoT systems, and intelligent networked systems. Experienced in developing AI-driven solutions for resource-constrained and real-world environments, with research spanning UAV-assisted data collection, network security, split learning, neural networks, and optimization. Proficient in Python, C/C++, PyTorch, TensorFlow, and modern AI/ML frameworks, with peer-reviewed publications in mobile computing, distributed systems, and intelligent systems. Interested in applying machine learning, deep learning, and autonomous AI technologies to challenging problems in intelligent and cyber-physical systems.

Education

Missouri University of Science and Technology, USA Aug 2023 – Present

Degree :	Ph.D. Student (GPA: 3.57 / 4) in Computer Science
Research:	UAV Networks, IoT systems, AI-driven data collection, distributed learning

University of Tabriz Aug 2016 - Jan 2019

Degree:	M.S. in Information Technology- Secure Communication (GPA: 3.4 / 4.0)
Research:	Mitigating Malicious Nodes in Multi-Channel Multi-Radio Mobile Networks

Technical Skills

Programming Languages**:**	Python, C/C++, OpenMP, CUDA, MPI, MapReduce , C#, Visual Basic
AI/ Machine Learning :	Deep Learning (CNN, RNN, GAN), Reinforcement learning, LLMs, NLP, Split learning
LLM & Agent Systems:	LangChain, Google ADK, N8N, Prompt Engineering, Workflow Automation
Frameworks & Libraries:	PyTorch, TensorFlow, Keras, Scikit-learn
Systems & Simulation Tools:	NS3, OMNeT ++, AERPAW
Technologies:	Docker, Git, Linux, LaTeX, Streamlit
Web & Database:	ASP.Net, C#, JavaScript, CSS, HTML, SQL, PostgreSQL, Snowflake

Experience

Graduate Research Assistant Aug 2023 - Present

Missouri University of Science and Technology Rolla, Mo

- Designed algorithms for detecting time-delay attacks in UAV networks
- Developed AI-based data collection frameworks for IoT systems
- Built and optimized neural network models for engineering applications
- Worked on distributed and multi-hop communication systems

Data Scientist & AI Specialist (ERDC Project) Jan 2024 - Dec 2025

Missouri University of Science and Technology Rolla, Mo

- Developed a production-oriented AI system for optimizing 3D-printed concrete
- Designed and deployed a Streamlit-based web application for real-time predictions
- Integrated machine learning models into user-facing applications
- Enabled non-technical users to interact with AI models through intuitive UI

Graduate Teaching Assistant Jan 2026 – May 2026

Missouri University of Science and Technology

- Instructed MATLAB programming
- Assisted students with computational problem solving

Graduate Assistant — Writing and Communication Center
Missouri University of Science and Technology Jun 2026 – Aug 2026

- Developed a comprehensive LaTeX dissertation template and technical user guide to streamline dissertation preparation for graduate researchers, supporting multi-paper integration, automated citations and cross-references, figures, tables, algorithms, and consistent document formatting.
- Mentored graduate and undergraduate students through individualized consultations on research papers, theses, dissertations, technical writing, and scholarly communication, translating complex technical and research concepts into clear, structured documentation.
- Coached international graduate students preparing for the university's Graduate Teaching Assistant oral communication assessment, delivering one-on-one training in technical presentation, instructional communication, and professional speaking.

Full Stack Developer Jan 2017 - Sep 2018

Industry experience

- Built backend APIs and web applications using ASP.NET, C#, and SQL
- Developed full-stack systems including frontend, backend, and database

Projects

Custom Autonomous Drone Platform

- Configured the hardware and software stack for autonomous flight and UAV research, including flight-controller setup, radio calibration, and system integration.
- Designed, assembled, and calibrated a quadcopter from scratch by integrating the flight controller, ESCs, motors, receiver, power system, and onboard computing.

AI Agent Automation Platform (Google ADK, LangChain)

- Designed and implemented multiple AI agents with tool-calling capabilities
- Built orchestration workflows (sequential and parallel) for multi-step reasoning
- Integrated external APIs for automated task execution
- Implemented structured inputs/outputs and session management

Automated AI Workflow System (Docker, n8n, Telegram)

- Built a fully automated pipeline to generate daily financial reports (gold, silver, Bitcoin, crypto)
- Integrated multiple APIs and automated workflows using n8n
- Deployed using Docker for reliable and continuous execution
- Delivered outputs automatically to a Telegram channel

AI Web Application (Streamlit)

- Designed and implemented an interactive web interface for AI-based predictions
- Integrated backend ML models with a real-time user-facing application
- Deployed online for accessibility and usability

Certificates

- Building AI Agents with Google ADK
- Designing Agentic Systems with LangChain
- LangGraph Essentials - Python
- Fundamentals of Deep Learning (NVIDIA)
- Azure Data Scientist Associate: Machine Learning
- Machine Learning (Standford University)
- Pandas (Kaggle)
- Associate Data Engineer in SQL
- Supervised Machine Learning: Regression and Classification (Standford University)

- Azure Data Scientist Associate: Machine Learning Services
- Learning Python (LinkedIn)
- Writing Efficient Python Code

Publications (Selected)

- Soltani et al., *ICDCN 2026* – Time-delay attack detection
- Soltani et al., *IEEE MASS 2025* – Energy-efficient split learning
- Soltani et al., *IEEE MASS 2024* – UAV data collection optimization
- Soltani et al., *Soft Computing Journal (2023)*
- Multiple journal papers under review (AI, UAV, materials modeling)

Professional Activities

- Publicity & Web Chair — PerAgent 2027 Workshop (co-located with IEEE PerCom 2027)
- Publicity Chair — 4th International Workshop on Unmanned Aerial Vehicles and IoT (UAV-IoT 2026)
- Technical Program Committee Member — IEEE Workshops (CINS 2025, ICDCN 2026,ICDCN 2027)
- Reviewer — ACM, IEEE, Elsevier journals
- Reviewer — Cement & Concrete Composites, Materials and Structures, Urban Geography

Honors and Awards

- NSF Travel Grant (2026)
- NSF Travel Grant (2025)
- Full Scholarship (M.S.)
- Ranked 2nd in B.S. program
- National competition recognition (software engineering)

Languege Skills

- English: Fluent (IELTS: 7.5)
- Kurdish: Native, Mother tongue

Persian: Native

Azerbaijani: Native